

Test 6: response

Signal Space: Test 6 response prerequisite

Technical state: completed. Scientific classification: pass.

Run: run-82cb109afa4e14c2. Analysis: analysis-0001-6a931d18.

Source profile run: run-080a63d117abd84d; frozen omega_Q=0.900, Omega=0.41274991; local radius=0.1.

Solver revision: 852a0ccec691d8a3bc6ac62bdf40730f848b5b4d; renderer revision: 852a0ccec691d8a3bc6ac62bdf40730f848b5b4d.

The exact locked method, thresholds and controls are in the plan. Scope: flat spherical matter; no gravitational or nonspherical result.

Locked scientific checks:

- amplitude-square: pass
- common-markers: pass
- conservation: pass
- counter-match: pass
- frozen-inputs: pass
- numerical-controls: pass
- prediction-first: pass
- response-match: pass
- sign-even: pass

packet-history

Question: Where do the compact neutral packets travel relative to the clock core?

Reading: Saved radial neutral-field histories show incoming propagation, central reflection and the separately prepared outgoing control. The upstream sphere is $r=14$.

Significance: This locates the forcing in space and time and makes the concentric-shell geometry explicit.

Limitation: Spherical focusing is physical in this restricted preparation; it is not an opposing-beam or recoil experiment. Color scales are independent.

upstream-waveform

Question: What neutral waveform actually reaches the upstream recording sphere?

Reading: The saved trace includes both initial passage and later returning radiation. Dividing by signed A overlays the weak single-packet waveforms.

Significance: Reception is compared against a recorded input field, with compact preparations and sign controls available for audit.

Limitation: A single scalar waveform does not separate incoming/outgoing characteristics; this run predicts from full registered initial data, not waveform-only inversion.

prediction-response

Question: Does the precomputed second-order response predict the local clock difference?

Reading: Base relative L2 prediction error is 0.00194 for one packet and 0.00193 for counterpropagating shells. Prediction files were locked before receiver runs.

Significance: Agreement tests the derived neutral-to-core-to-clock response coefficient, beyond the symmetry-enforced sign parity.

Test 6: response

Limitation: The Taylor predictor and nonlinear solver share the discrete Hamiltonian; independent Hamiltonian and Taylor checks plus numerical refinements constrain, but do not eliminate, shared implementation error.

parity-scaling

Question: Is the clock difference even in packet sign and quadratic in amplitude?

Reading: Weak amplitude exponent 1.9993540; normalized ladder difference 0.00728; sign-odd/even norm ratio 0.

Significance: The amplitude ladder discriminates second-order forcing from a linear response, while sign parity checks the exact action symmetry.

Limitation: Exact numerical sign parity is strongly enforced by this deterministic action. It alone does not validate the magnitude or operational meaning of the clock response.

marker-convergence

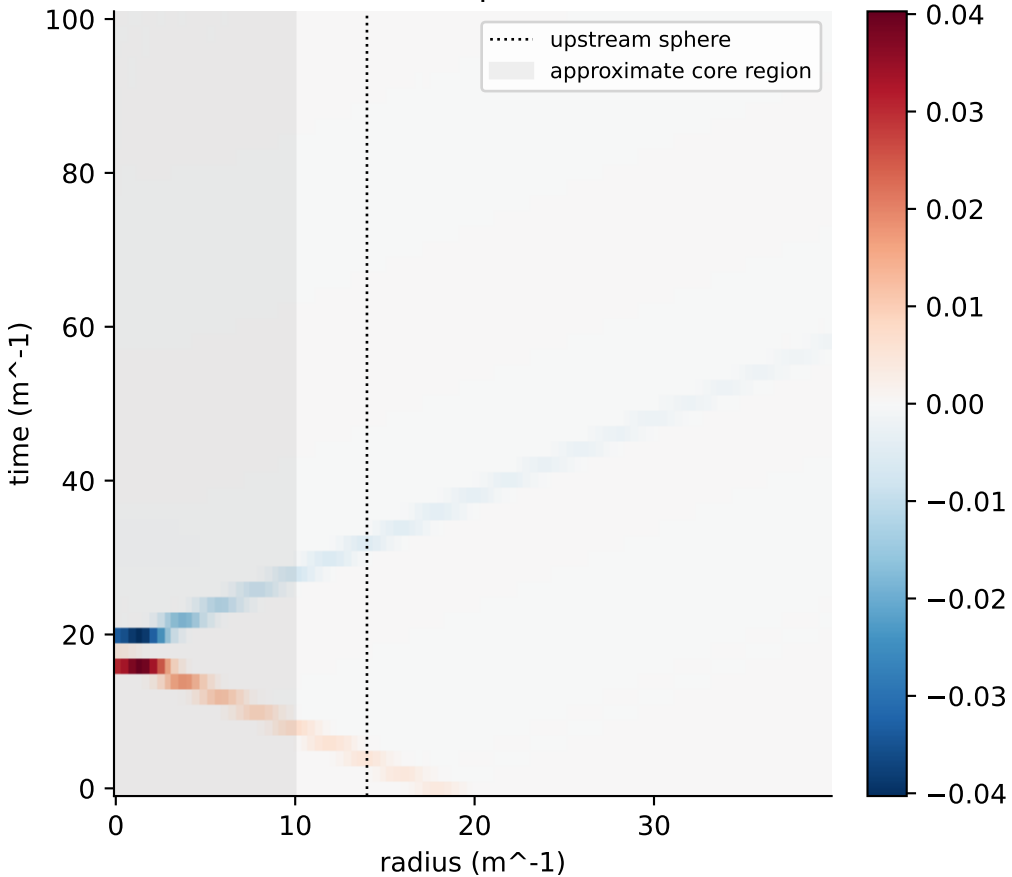
Question: Are corresponding local tick shifts reproducible under mesh, time and box controls?

Reading: Every base preparation has 6 baseline ticks available for pairing. Largest base tick shift $5.32\text{e-}07$ cycles; largest predicted tick-shift error $8.42\text{e-}09$ cycles.

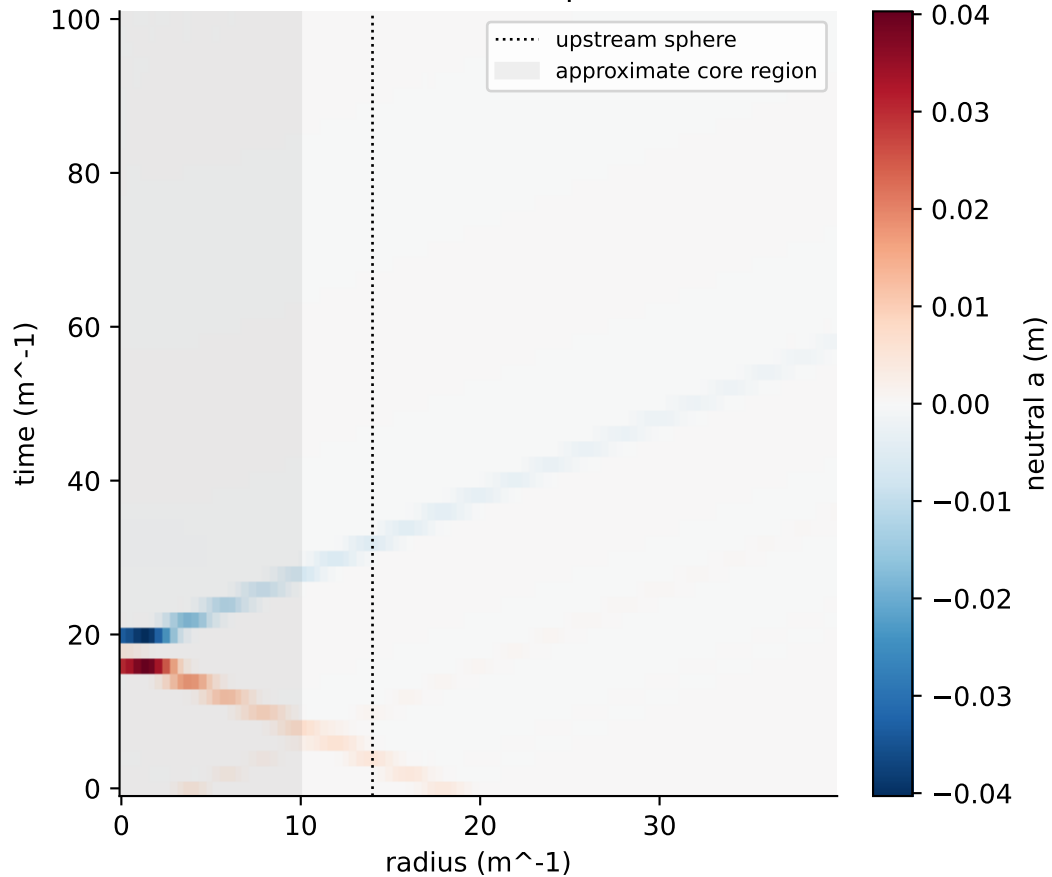
Significance: The measured response reaches an identifiable local clock record, with numerical sensitivity displayed independently.

Limitation: Tick ordinals are anchored to the common preparation in one flat frame. Physical emission/reception marker invariance and two-object recoil are still untested. Values below $1\text{e-}16$ are plotted at that floor.

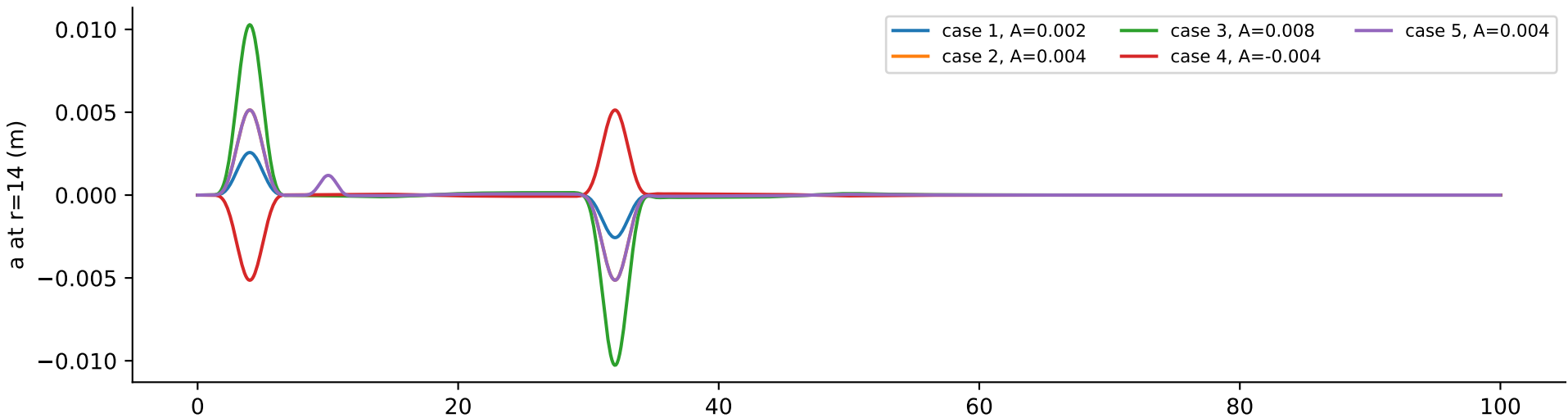
Inward packet



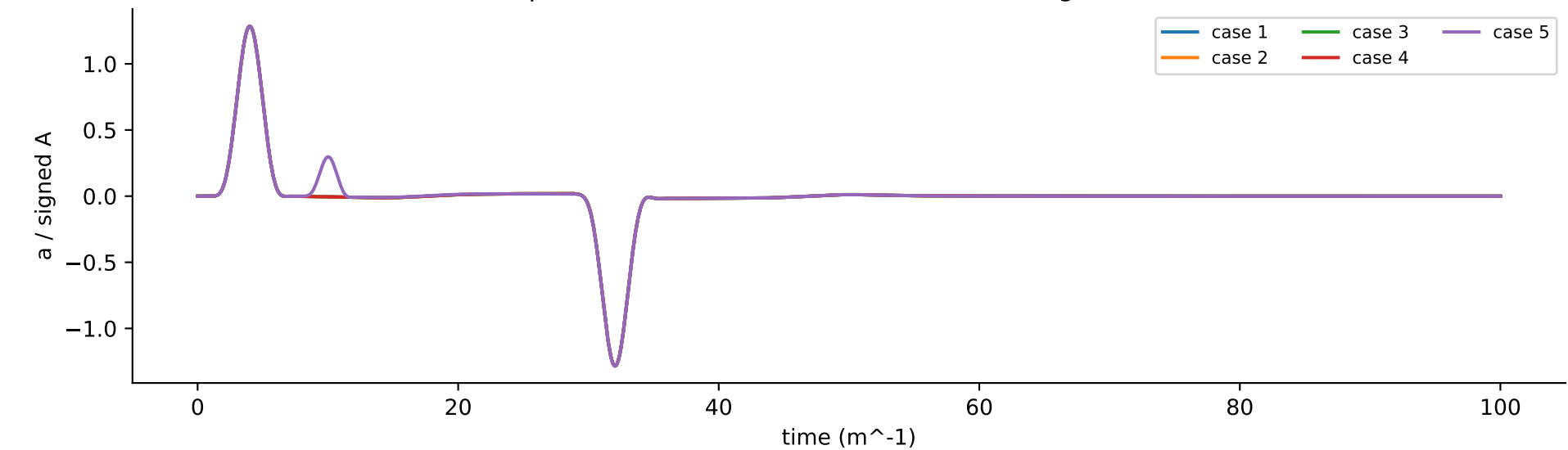
Inward + outward packets

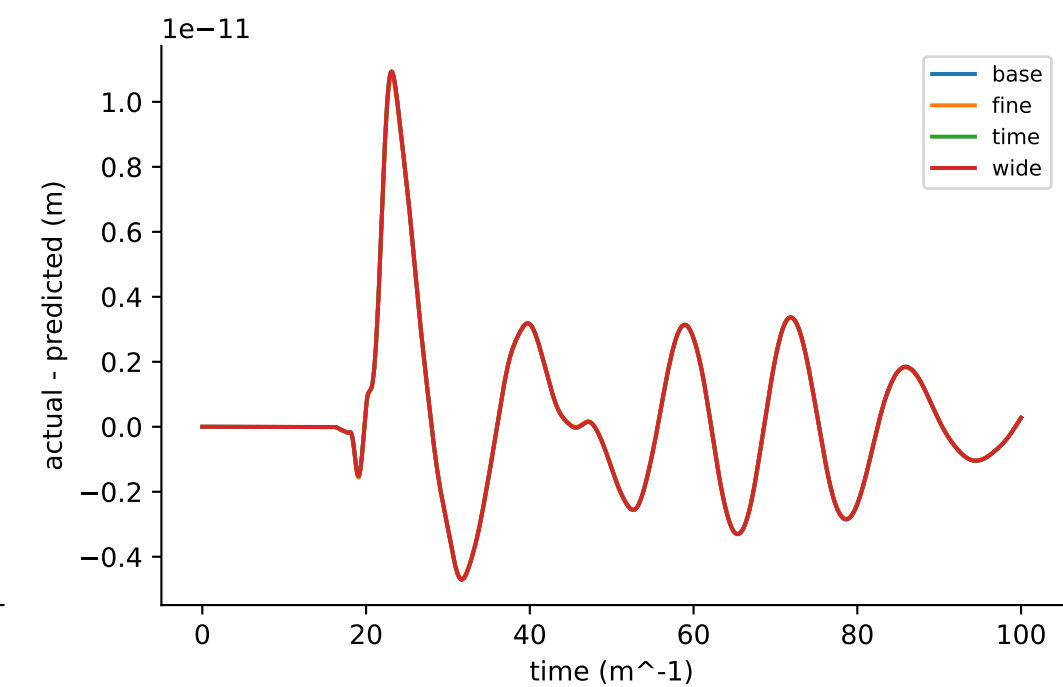
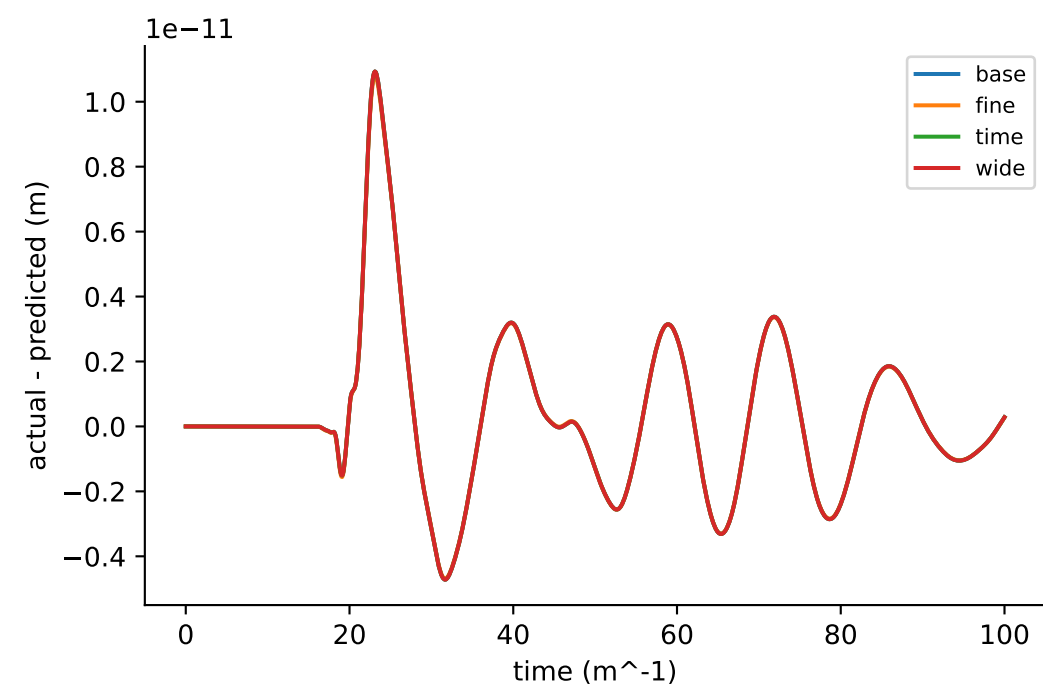
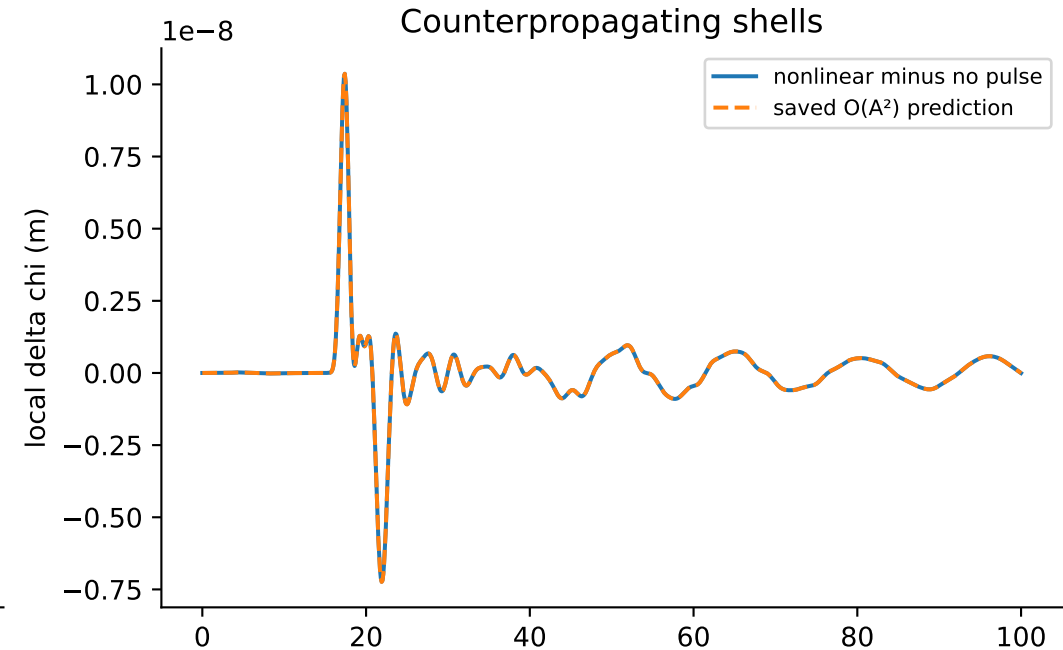
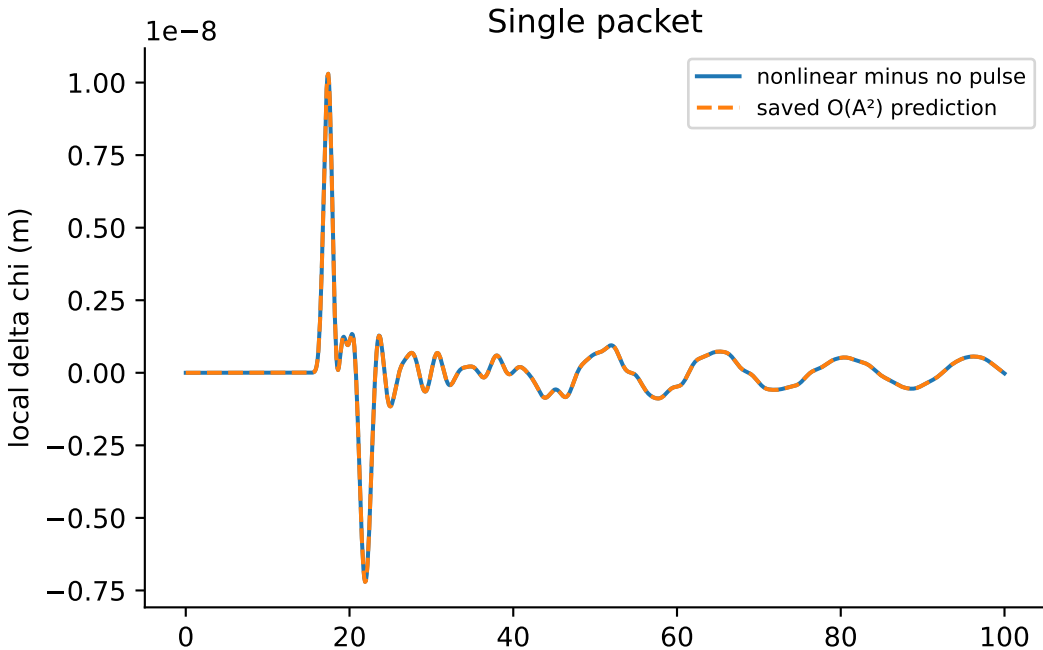


Actual recorded upstream field



Amplitude-normalized incident and returning waveform





Weak response slope = 1.999354

